Copernicus Global Land Operations
”Vegetation and Energy”
“CGLOPS-1”
Framework Service Contract N° 945120-IPR-2023

PRODUCT USER MANUAL

Leaf Area Index (LAI)
Fraction of Absorbed Photosynthetically Active Radiation (FAPAR)
Fraction of green Vegetation Cover (FCover)
Version 2.0

Issue I1.00

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Dissemination Level		
PU	Public	X
PP	Restricted to other programme participants (including the Commission Services)	
RE	Restricted to a group specified by the consortium (including the Commission Services)	
CO	Confidential, only for members of the consortium (including the Commission Services)	

Document Release Sheet

Book captain:	Dominique de Munck (VITO)	Sign *Dominique De Munck*	Date 17.12.2025
Approval:	Roselyne Lacaze (HYGEOS)	Sign *Roselyne Lacaze*	Date 19.12.2025
Endorsement:	N. Gobron (JRC)	Sign	Date
Distribution:	Public		

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Change Record

Date	Page(s)	Description of Change	Release
17.12.2025	All	First issue for Version 2.0 of Sentinel-3 and PROBA-V products	I1.00

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List of Acronyms

AC	Atmospheric Correction
AD	Applicable Document
ATBD	Algorithm Theoretical Basis Document
AVHRR	Advanced Very-High-Resolution Radiometer
B	Mean Bias
BELMANIP	BEnchmark Land Multisite ANalysis and Intercomparison of Products
CCI-LC	Climate Change Initiative-Land Cover
CEOS	Committee on Earth Observation Satellite
CGLOPS	Copernicus Global Land OPerationS
CREAF	Centre for Ecological Research and Forestry Applications
CLMS	Copernicus Land Monitoring Service
DN	Digital Number
EBF	Evergreen Broadleaf Forest
EPS	EUMETSAT Polar System
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
FAPAR	Fraction of Absorbed Photosynthetically Active Radiation
FCover	Fraction of Vegetation Cover
FIPAR	Fraction of Intercepted Photosynthetically Active Radiation
FP7	7th Framework Programme
JRC	Joint Research Center
GAUL	Global Administrative Unit Layer
GBOV	Ground Based Observations for Validation
GCOS	Global Climate Observing System
GeoTIFF	Geographic Tagged Image File Format
GLASS	Global LAnd Surface Satellite
GLOBCOVER	Global land cover from MERIS
IDEPIX	Identification of Pixels processor
ImagineS	Implementation of Multi-scale Agricultural Indicators Exploiting Sentinels
LAI	Leaf Area Index

LANDVAL	Land Validation network of sites
LPV	Land Product Validation
LSA SAF	Land Surface Analysis Satellite Applications Facility
MERIS	Medium Resolution Imaging Spectrometer
NetCDF-CF	Network Common Data Form Climate and Forecast metadata convention
NASA	National Aeronautics and Space Agency
NIR	Near Infrared
NNT	Neural Network Technique
NOBS	Number of observations
OLCI	Ocean and Land Colour Instrument
PROBA-V	VEGETATION sensor on Project for OnBoard Autonomy platform
PUM	Product User Manual
QA	Quality Assessment indicators
QAR	Quality Assessment Report
QC	Qualitative quality control indicators
QFLAG	Quality Flag
R	Correlation coefficient
RMSD	Root Mean Square Deviation
RMSE	Root Mean Square Error
RT	Real Time
S3	Sentinel-3
SAA	Sun Azimuth Angle
SLSTR	Sea and Land Surface Temperature Radiometer
SMAC	Simplified Method for Atmospheric Correction
SVAT	Soil-Vegetation-Atmosphere Transfer model
SZA	Sun Zenith Angle
TOC	Top Of Canopy
VAA	View Azimuth Angle
VEGA	LSA SAF EPS-AVHR vegetation products
VIIRS	NASA Visible/Infrared Imager Radiometer Suite
VITO	Vlaamse Instelling voor Technologisch Onderzoek (Flemish Institute for Technological Research), Belgium

VNIR	Visible and Near Infrared
VZA	View Zenith Angle
WGS84	World Geodetic System 1984
XML	Extensible Markup Language
XSL	Extensible Stylesheet Language

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EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery of the parameters take place in a timely manner and are complemented by the constitution of long-term time series.

The CLMS delivers the Leaf Area Index (LAI), the Fraction of Absorbed Photosynthetically Active Radiation absorbed by the vegetation (FAPAR) and the Fraction of green Vegetation Cover (FCover) at 300m resolution. The Version 1.0 of these products, produced and delivered in near Real Time (RT), were derived from PROBA-V Collection 1 data from January 2014 to June 2020, end of the operational PROBA-V mission. The Version 1.1. of the algorithm has been adapted to the new collection of PROBA-V data, Collection 2, and to the imagery of the Ocean and Land Colour Instrument (OLCI) onboard the Sentinel-3 platform. These Version 1.1 Sentinel-3-based products are delivered from July 2020 ensuring the continuity of PROBA-V 300m LAI, FAPAR, and FCover products. Some marginal inconsistencies were identified between PROBA-V and Sentinel-3 300m products due to the differences between Version 1.0 and Version 1.1 retrieval algorithms and input auxiliary data, and because of some issues in the implementation of Version 1.0 and Version 1.1 algorithms. Further, the Collection 2 of PROBA-V data has been released in March 2023. To improve the consistency of the long time series of LAI, FAPAR and FCover products at 300m resolution from 2014 to present, we have adapted Version 1.1 to the characteristics of PROBA-V Collection 2 input data and to the Version 2.3 of Sentinel-3 OLCI surface reflectance for the reprocessing of the entire time series, from PROBA-V and Sentinel-3, resulting in Version 2.0 algorithm.

This Product User Manual (PUM) describes the CLMS LAI, FAPAR, and FCover 300m Version 2.0. The variables are calculated globally on a 10-day basis and made available to the user in near-real time every 10 days. The product is projected on a regular latitude/longitude grid with a resolution of 1/336°. It is delivered covering the whole globe (from -180°E to +180°W and from +85°N to -65°S) and provided in netCDF4-CF format (by default format) containing the variables values (LAI, FAPAR or FCover), associated with some quantitative (QA) and qualitative (QC) quality indicators.

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

The PUM is the primary document that users have to read before handling the products.

It includes a summary of the retrieval methodology and of main validation results and it provides detailed information on file naming, file format, product content, product characteristics, data policies and data access.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 summarizes the retrieval methodology.
- Chapter 3 describes the technical characteristics of the product.
- Chapter 4 reports the main results of the quality assessment.
- Chapter 5 presents some basic usage and gives recommendations on efficient use of products

Users' requirements are recalled in Annex.

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2020/OP/3213, 17th December 2020.

Available at https://land.copernicus.eu/global/documents/applicable

1.3.2 Input

Document ID	Descriptor
CGLOPS1_PQAD	Product Quality Assurance Document of the Copernicus Global Land Service for "Vegetation & Energy" produts.
CGLOPS1_ATBD_ LAI[FAPAR/FCOVER]300m-V2.0	Algorithm Theoretical Basis Document of LAI, FAPAR, FCOVER 300m Version 2.0

CGLOPS1_VR_ LAI[FAPAR/FCOVER]300m-V2.0	Validation Report of the LAI, FAPAR, FCover 300m Version 2.0 product
CGLOPS1_ATBD_S3-AC-V1.1	Algorithm Theoretical Basis Document of the atmospheric corrections applied on the Sentinel-3 data
CGLOPS1_PUM_S3-TOC-Reflectance300m-V2.3	Product User Manual of Sentinel-3 Top of Canopy (TOC) reflectance 300m V2.3
CGLOPS1_QAR_S3-CloudMask	Report presenting the evaluation of OLCI and SLSTR cloud, cloud shadow and snow masks.
CGLOPS1_QAR_LAI[FAPAR/FCOVER]300m-V1	Quality Assessment Report of the PROBA-V 300m LAI, FAPAR, FCOVER Version 1.0 products

These documents are available on the CLMS website: https://land.copernicus.eu/en/technical-library

1.3.3 External documents

Document ID	Descriptor
ImagineS_RP2.1_ATBD-LAI300m_I1.73	Algorithm Theoretical Basis Document describing the Version 1.0 of retrieval methodology of LAI, FAPAR, FCover 300m from PROBA-V

Available at: https://land.copernicus.eu/global/documents/products

PROBAV_PUM_C2 PROBA-V Collection 2 Products User Manual v1.0, 27 March 2023

Available at https://proba-v.vgt.vito.be/sites/probavvgt/files/downloads/PROBA-V_C2_Products_User_Manual.pdf

ACRI-ST, 2017 Product Data Format Specification – OLCI Level 1 products

Available at: https://sentinel.esa.int/documents/247904/1872756/Sentinel-3-OLCI-Product-Data-Format-Specification-OLCI-Level-1

GCOS#245 The 2022 GCOS ECVs Requirements

https://library.wmo.int/opac/doc_num.php?explnum_id=3417

2 ALGORITHM

2.1 INTRODUCTION

Vegetation state and dynamics play a key role in global climate-carbon cycle. The importance of continuously monitoring the Earth's surface was recognized by Global Climate Observing System. A set of Essential Climate Variables was identified as being both accessible from remote sensing observations and intervening within key processes (GCOS#245). Among those related to land surfaces, the leaf area index (LAI) and the fraction of absorbed photosynthetic active radiation (FAPAR) may be derived from observations in the reflective solar domain. These vegetation biophysical variables are crucial in several processes, including photosynthesis, respiration and transpiration. In addition to LAI and FAPAR, the fraction of green vegetation (FCover) as seen from nadir appears also as a very pertinent variable for partitioning contributions between soil and vegetation in land surface models.

2.2 DEFINITION OF THE VARIABLES

LAI is defined as one half the total green (i.e., photosynthetically active) leaf area per unit horizontal ground surface area (Chen and Black 1992). LAI is a non-dimensional quantity, although units of m^2/m^2 are often quoted. It determines the size of the interface for exchange of energy (including radiation) and mass between the canopy and the atmosphere. This is an intrinsic canopy primary variable that should not depend on observation conditions. LAI is strongly non-linearly related to reflectance. Therefore, its estimation from remote sensing observations is scale dependent (Garrigues et al. 2006; Weiss et al. 2000). Note that vegetation LAI as estimated from remote sensing includes all the green contributors such as the understory when existing under forests canopies.

FAPAR is defined as the fraction of photosynthetically active radiation (PAR; solar radiation reaching the surface in the 0.4-0.7µm spectral region) that is absorbed by vegetation. FAPAR is expressed as a unitless fraction of the incoming radiation received at the land surface. The FAPAR value results directly from the radiative transfer in the canopy. FAPAR is the sum of two terms, weighted by the diffuse fraction in the PAR domain: the 'black sky' fAPAR that corresponds to the direct component (collimated beam irradiance in the sun direction only) and the 'white sky' or the diffuse component. FAPAR can be computed at a given time (e.g instantaneous fAPAR at the actual sun position of measurement) or daily integrated. It depends on canopy structure, vegetation element optical properties and illumination conditions. FAPAR is very useful as input to a number of primary productivity models based on simple efficiency considerations (McCallum et al. 2009; Prince 1991). Most of the primary productivity models using this efficiency concept are running at the daily time step. Since the CLMS FAPAR product is originally derived from CYCLOPES FAPAR at 10:00 (Baret et al. 2007) and MODIS aboard Terra FAPAR at 10:30 (Myneni et al. 2002), CLMS FAPAR product is defined as the black-sky fraction of PAR absorbed by green elements including over and

understorey vegetation at 10:15 (actually between 10:00 and 10:30) which is a good approximation of the daily integrated black-sky FAPAR value (Baret et al. 2007). FAPAR is relatively linearly related to reflectance values, and is little sensitive to scaling issues (Hilker et al. 2010; Weiss et al. 2000). Note also that the FAPAR refers only to the green parts of the canopy.

FCover is defined as the fraction of ground surface covered by green vegetation as seen from the nadir direction (Baret et al. 2013). This definition agrees with the definition of LSA SAF FCover products (https://landsaf.ipma.pt/en/products/vegetation/fvc/) as well as the GLASS FCover product (Jia et al. 2015). FCover is expressed as a unitless fraction of the ground surface. FCover is used to separate vegetation and soil in energy balance processes, including temperature and evapotranspiration. It is computed from the leaf area index and other canopy structural variables and does not depend on variables such as the geometry of illumination as compared to FAPAR. For this reason, it is a very good candidate for the replacement of classical vegetation indices for the monitoring of green vegetation. Because of its quasi-linear relationship with reflectance, FCover is only marginally scale dependent (Weiss et al. 2000). Note that similarly to LAI and FAPAR, only the green elements are considered.

2.3 PROBA-V INSTRUMENTS AND DATA

The PROBA-V sensor has been launched on 6th May 2013 onboard the PROBA platform. It was designed to bridge the gap in space-borne vegetation measurements between SPOT-VGT (March 2018 – May 2014) and the Sentinel-3 satellites launched in 2016. The mission objective was to ensure the continuity with the heritage of the SPOT-VGT mission.

PROBA-V operated at an altitude of 820 km altitude in a sun-synchronous orbit with a local overpass time at launch of 10:45 h. Because the satellite had no onboard propellant, the overpass time was expected to gradually differ from the at-launch value. After launch, the local overpass time first increased to 10:50h in October 2014, followed by a decrease to 10:45h in June 2016. By end-of-mission in June 2020, the Local Time of Descending Node will be at ~09:30h.

The instrument had a Field of View of 102°, resulting in a swath width of 2295 km. This swath width ensured a daily near-global coverage (90%) and full global coverage is achieved every 2 days. An array of 6000x4 elements was used in the VIS-NIR (only 3 bands on the 4 potential ones are used) yielding to a ground sampling distance that varied across the swath from 100m up to 350m at the extremities of the swath. The SWIR domain was sampled using 3 arrays of 1024 elements, providing a ground sampling distance about twice as that in the VIS-NIR.

Table 1: PROBA-V spectral characteristics: band center and width. The spectral bands selected for LAI, FAPAR, FCover 300m algorithm are highlighted in bold.

Band	Center (nm)	Width (nm)	Potential Applications
B0 (blue)	**463**	**46**	**Continental ecosystems - Atmosphere**
B2 (red)	**655**	**79**	**Continental ecosystems**
B3 (NIR)	**845**	**144**	**Continental ecosystems**
SWIR	1600	73	Continental ecosystems

2.4 SENTINEL-3 INSTRUMENTS AND DATA

Sentinel-3 is a constellation of, at least, 2 identical satellites, Sentinel-3A and 3B. Sentinel-3A was launched on 16th February 2016 followed by the launch of Sentinel-3B on 25th April 2018. Each satellite operates in a reference orbit with a repeat cycle of 27 days, with a 4-day sub-cycle. The use of Sentinel-3A and Sentinel-3B satellites in conjunction enable a short revisit time of less than two days at the equator. Satellites fly at 814 km altitude on a circular sun-synchronous orbit with 10:00 a.m. equatorial crossing time.

The Sentinel-3 platforms have on board the optical sensor OLCI with 300m spatial resolution, swath=1269 km, field of view 68° and 21 spectral channels. The OLCI benefits from MERIS heritage. The field-of-view of OLCI is divided between five cameras on a common structure with the calibration assembly. Each camera has an optical grating to provide the minimum baseline of 16 spectral bands required by the mission together with the potential for optional bands for improved atmospheric corrections (Table 2). As for MERIS, OLCI was preliminary designed for ocean applications. However, considering its spectral characteristics and spatial resolution, it is fully suitable for land applications.

Table 2: Sentinel-3 OLCI spectral characteristics: band center and width. White lines correspond to Medium Resolution Imaging Spectrometer (MERIS) heritage while grey lines correspond to OLCI additional bands. The spectral bands selected for LAI, FAPAR, FCover 300m algorithm are highlighted in bold.

Band	λ centre (nm)	Width (nm)	Potential Application
Oa1	400	15	Aerosol correction, improved water constituent retrieval
Oa2	412.5	10	Yellow substance and detrital pigments (turbidity)
Oa3	442.5	10	Chl absorption max., biogeochemistry, vegetation
Oa4	**490**	**10**	**High Chlorophyll, other pigments**
Oa5	**510**	**10**	**Chlorophyll, sediment, turbidity, red tide**
Oa6	**560**	**10**	**Chlorophyll reference (Chlorophyll minimum)**
Oa7	**620**	**10**	**Sediment loading**
Oa8	**665**	**10**	**Chlorophyll (2nd Chlorophyll abs. max.), sediment, yellow substance /vegetation**
Oa9	**673.75**	**7.5**	**For improved fluorescence retrieval and to better account for smile together with the bands 665 and 680 nm**
Oa10	**681.25**	**7.5**	**Chlorophyll fluorescence peak, red edge**
Oa11	**708.75**	**10**	**Chlorophyll fluorescence baseline, red edge transition**
Oa12	**753.75**	**7.5**	**Oxygen absorption/clouds, vegetation**
Oa13	761.25	2.5	Oxygen absorption band/aerosol correction
Oa14	764.375	3.75	Atmospheric correction
Oa15	767.5	2.5	Oxygen absorption band used for cloud top pressure, fluorescence over land
Oa16	**778.75**	**15**	**Atmospheric correction/aerosol correction**
Oa17	**865**	**20**	**Atmospheric correction/aerosol correction, clouds, pixel co-registration**
Oa18	**885**	**10**	**Water vapour absorption reference band. Common reference**
Oa19	900	10	Water vapour absorption, vegetation monitoring
Oa20	940	20	Water vapour absorption, atmos./aerosol corr.
Oa21	1 020	40	Atmospheric correction/aerosol correction

2.5 RETRIEVAL METHODOLOGY

2.5.1 Background

The objective is to develop an algorithm dedicated to the estimation of Version 2.0 LAI, FAPAR and FCover 300m from the PROBA-V and Sentinel-3 series of observations. The algorithm should provide high level of consistency with current Version 1.0 of PROBA-V products. The Version 2.0 of LAI, FAPAR and FCover 300m products should have the same temporal resolution of 10 days. Products should also be associated with quality assessment flags as well as quantified uncertainties. The algorithm runs at the pixel level without interactions with the surrounding pixels. The algorithm should provide real time estimation. This forces to perform short term projection of the product dynamics.

Version 1.0 of 300m product retrieval algorithm was set-up in the ImagineS project [ImagineS_RP2.1_ATBD-LAI300m_I1.73]. After the quality assessment of the 300m LAI, FAPAR, FCover Version 1.0 [CGLOPS1_QAR_LAI[FAPAR/FCOVER]300m-V1], the methodology has been adapted to reduce the noise in the product time series especially in near real time mode and to address external recommendations and operational constraints in CLMS. Since the performance of LAI, FAPAR, FCover estimates is highly dependent on the level of noise in the input reflectance data, Top of Atmosphere (TOA) reflectance used in Version 1.0 have been replaced by TOC reflectance in the Version 2.0 of the algorithm. Further, the parametrization of the algorithm has been adapted in Version 2.0 to the characteristics of PROBA-V Collection 2 and Sentinel-3 Version 2.3 TOC reflectance data. The neural networks used to transform PROBA-V and Sentinel-3 TOC reflectance into instantaneous LAI, FAPAR and FCover estimates have been calibrated with current Version 1.0 PROBA-V 300m LAI, FAPAR, FCover products to ensure a good agreement and temporal consistency in the transition from PROBA-V to Sentinel-3. Details of the Version 2.0 LAI, FAPAR and FCover 300m are provided in Section 2.5.4. The differences with previous versions of 300m are detailed in Section 2.6.

2.5.2 Outline

The scheme proposed for the retrieval methodology is sketched in Figure 1. The algorithm starts from instantaneous PROBA-V and Sentinel-3 TOC reflectance products which are first transformed into instantaneous estimates of LAI, FAPAR, FCover (Step A in Figure 1). Then, smoothing and gap filling is achieved over a compositing temporal window that may be dissymmetric as in the case of the near-real time situation or at the beginning of the time series (Step B in Figure 1).

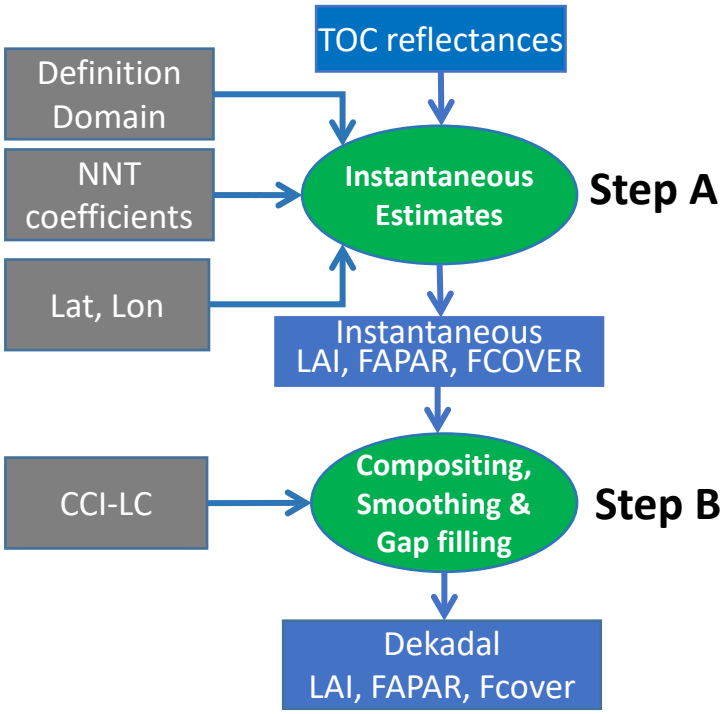


Figure 1: Flow chart showing the two processing steps in LAI, FAPAR, FCover 300m algorithm.

Two different processing chains are applied whether the pixel's biome is an Evergreen Broadleaf Forests (EBF) or not (nonEBF). Gap filling is only applied for EBFs.

The selection of the instantaneous estimates considered as valid in the compositing window (Step B) is made based on the LAI product.

The compositing for real time estimates (Step B) is achieved according to the scheme described in Figure 2. We consider a particular dekadal date, *D*.

- In real time mode, RT0 (when the actual date corresponds to *D*, i.e. the first line in Figure 2), the compositing is achieved using only the past observations, with a maximum compositing window spanning *c-* days in the past (*c-* was set to 210 days for EBF and 60 days for nonEBF). The value of the product at dekadal date *D* is therefore relatively unstable in the presence of noisy and missing data.
- Then, the observations accumulate after dekadal date *D* when time passing. The value at dekadal date *D* is updated using the observations available after dekadal date *D* within the consolidation period. The accuracy of the product value progressively improves during this consolidation period.

- At the end of the consolidation period, the value of the product at dekadal date *D* has converged towards the 'historical' time series when, for a given dekad 'D' to be processed, the 'n' dekads before 'D' and after 'D' in the time series are available. 'n' is the number of dekads required for the convergence of LAI, FAPAR and FCover values. 'n' was fixed to 6 dekads, i.e. 60 days.

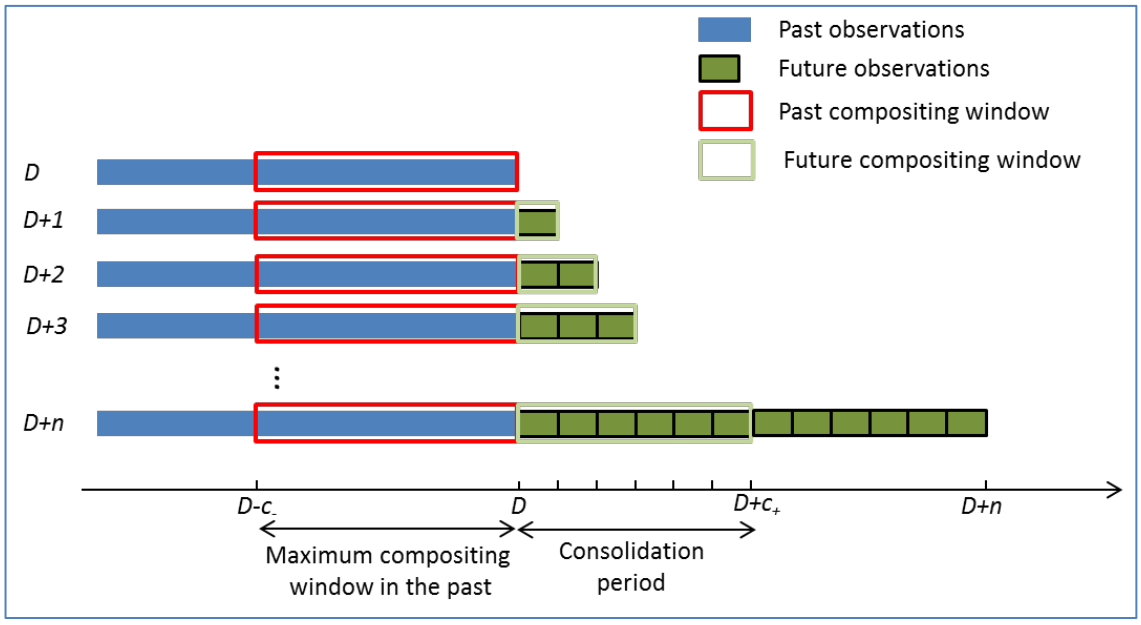


Figure 2: Scheme showing the compositing used for near real time estimates from RT0 (in the top) to the final consolidation RT6 (in the bottom). D refers to the dekad being processed. D+n corresponds to the n dekads available after the dekad D. c- refers to the maximum semi-compositing window before date D (60 days for nonEBF and 210 days for EBF) and c+ refers to the semi-compositing window after date D (0 days for RT0, 10 days for RT1, 20 days for RT2, 30 days for RT3, and 60 days for RT6 and after if the historical time series are reprocessed).

Note that when working in real time estimation (RT0) and updating the values for each new dekad available (RT1 and RT2), the compositing period is not symmetric: the maximum compositing window in the past before date *D* is longer than the maximum compositing window after date *D*: $c_- > c_+$. The compositing period after D, c_+, is limited by the length of the consolidation period. For nonEBFs in RT6 case or when the time series are reprocessed in 'historical' mode after *D+6* dekads, the maximum length of the compositing period after date *D* should be equal to that in the period before, i.e. $c_+ = c_- = 60$ days. However, for EBFs, the compositing window is never symmetric: $c_- = 210$ days and $c_+ = [0,60]$ days.

Once a new dekad is available after dekadal date D, the value of the RT product at dekadal date D is recomputed. Since the product value after the second consolidation (RT2) remains mostly stable, the third, fourth and fifth consolidations are not distributed but only RT0, RT1, RT2 and RT6, which is the final product. If the time series are reprocessed in 'historical' mode after *D+6* dekads, only RT6 products are produced and distributed.

2.5.3 Input data

All these inputs are required for each considered pixel.

2.5.3.1 Top of canopy reflectance

The input data of the 300m LAI, FAPAR, FCover Version 2.0 processing line are the Top of Canopy (TOC) reflectance from Collection 2 of PROBA-V in blue, red and near infrared spectral bands (B0. B2 and B3) [PROBAV_PUM_C2] and OLCI/Sentinel-3 in twelve Visible and Near Infrared (VNIR) spectral bands (Oa4-12 and Oa16-18, Table 2) [CGLOPS1_ATBD_S3-AC-V1.1, CGLOPS1_PUM_S3-TOC-Reflectance300m-V2.3].

2.5.3.2 Geometry of acquisition

The geometry information is required for the neural networks. It includes:

- the cosine of the view zenith angle (cos(VZA)),
- the cosine of the sun zenith angle (cos(SZA)),
- the cosine of the relative azimuth angle (cos(SAA-VAA)) where SAA corresponds to the Sun Azimuth Angle and VAA to the Viewing Azimuth Angle

2.5.3.3 Land cover

The CCI-LC (https://www.esa-landcover-cci.org) land cover map was used as auxiliary information for deciding whether a pixel belongs or not to an Evergreen Broadleaf Forest (EBF) at the beginning of the time series. The CCI-LC **(**2.0.7cds**)** classification for the first year of time series processing is used: i.e. CCI-LC 2014.

2.5.3.4 Land/water mask

A mask identifying both open ocean pixels and inland water pixels as "water" must be used as auxiliary information for deciding whether a pixel belongs to land or water. This mask is inherited from the quality flags of OLCI Sentinel-3 TOC reflectance. 'IDEPIX_LAND', from Idepix cloud detection module, and 'fresh_inland_water' flag, from OLCI L1B data, are used to distinguish "land" and "water" pixels. Only land pixels are processed. One flag (QC(1) in Table 6) associated to the LAI, FAPAR, FCover products specifies if the pixel is identified as land (QC(1)=1) or water (QC(1)=0).

2.5.4 Processing steps

The retrieval methodology relies on 2 main steps as shown in Figure 1:

1. Step A: estimate of instantaneous LAI, FAPAR, FCover values
2. Step B: compositing, filtering and gap filling (gap filling is only applied for EBFs processing).

The general principles are summarized below. More details can be found in the ATBD [CGLOPS1_ATBD_ LAI[FAPAR/FCOVER]300m-V2.0].

2.5.4.1 Step A: estimates of instantaneous LAI, FAPAR, FCover values

The input TOC reflectances are transformed into instantaneous LAI, FAPAR and FCover estimates as described in Figure 3.

1. ***Rejection of data based upon their status map (A1):*** The quality flags associated to PROBA-V and Sentinel-3 TOC data are first used to keep only pixels with all the selected spectral bands having good radiometric quality, located over land, not covered by ice, cloud or snow.
2. ***First outlier rejection based on inputs (A2):*** The algorithm verifies first whether the inputs of a given observation keep within the range of variation of the training dataset used for algorithm calibration. TOC reflectance which are outside the definition domain are removed. Note that the definition domain is sensor specific: one domain is defined for PROBA-V and one domain for Sentinel-3/OLCI.
3. ***Instantaneous estimates of LAI, FAPAR, FCover (A3)***. The top of the canopy reflectances are then transformed into instantaneous products using neural networks (NNT). Two sets of specific NNT for EBF and nonEBF were previously calibrated for each of the 3 variables considered (LAI, FAPAR, and FCover). The neural networks are also sensor specific: one set of neural networks is calibrated for PROBA-V and other for Sentinel-3/OLCI. Then, the combination of three variables (LAI, FAPAR, FCover) per two landcover vegetation classes (EBF and nonEBF) per two sensors (PROBA-V and Sentinel-3/OLCI) resulted in twelve neural networks. Since the objective was to achieve consistency with PROBA-V V1.0 LAI, FAPAR and FCover 300m products, the NNT were trained using concurrent PROBA-V or Sentinel-3/OLCI TOC reflectances and PROBA-V V1.0 LAI, FAPAR and FCover products in the overlapping period from 1/07/2018 to 30/06/2019 over the BELMANIP2.1 sites (Baret et al. 2006; Weiss et al. 2014).
   The neural networks architecture used in the Version 2.0 retrieval is exactly the same as the one proposed for the generation of Version 1.0 of 300m LAI, FAPAR and FCover products [CGLOPS1_ATBD_ LAI[FAPAR/FCOVER]300m-V2.0]. The NNT are composed of:

- One input layer made of the input TOC data in the selected spectral bands (the PROBA-V neural networks ingest TOC reflectance in 3 spectral bands: B0, B2 and B3, while the Sentinel-3/OLCI neural networks ingest TOC reflectance in 12 OLCI spectral bands: Oa4-12 and Oa16-18) and the cosine of the 3 associated sun-view angles (sun and view zenith angles, and relative azimuth angle).
- One hidden layer with 5 neurons having tangent sigmoid transfer functions.
- One output layer with a linear transfer function.

This NNT architecture is in good agreement with what proposed by Verger et al. (2011).

4. ***Outlier rejection based on output values (A4):*** when outputs of neural networks are out of the physical range of values (Table 4) extended by the tolerance limits ([-0.2, 10] for LAI, and [-0.1, 1.2] for FAPAR and FCover) for one (or more) of the three variables (LAI, FAPAR, FCover), they are considered invalid for all the three variables. This allows keeping a high level of consistency between the three variables. Values that are within the tolerance limits but higher (lower) than the physical maximum (minimum) are fixed to the physical maximum (minimum).
5. ***Ensuring FAPAR-FCover consistency (A5):*** The FAPAR value divided by its maximum theoretical value (0.94, Baret and Guyot (1991)) must remain higher than FCover in any situations since it can be approximated by the FIPAR (Fraction of Intercepted Photosynthetically Active Radiation). Therefore, to ensure FAPAR-FCover consistency, a final condition is applied:

$$FCover = \min\left(FCover_{NNT}, \frac{FAPAR_{NNT}}{0.94}\right)$$

Where $FCover_{NNT}$ and $FAPAR_{NNT}$ are the instantaneous estimates issued from the FCover and FAPAR NNT, respectively.

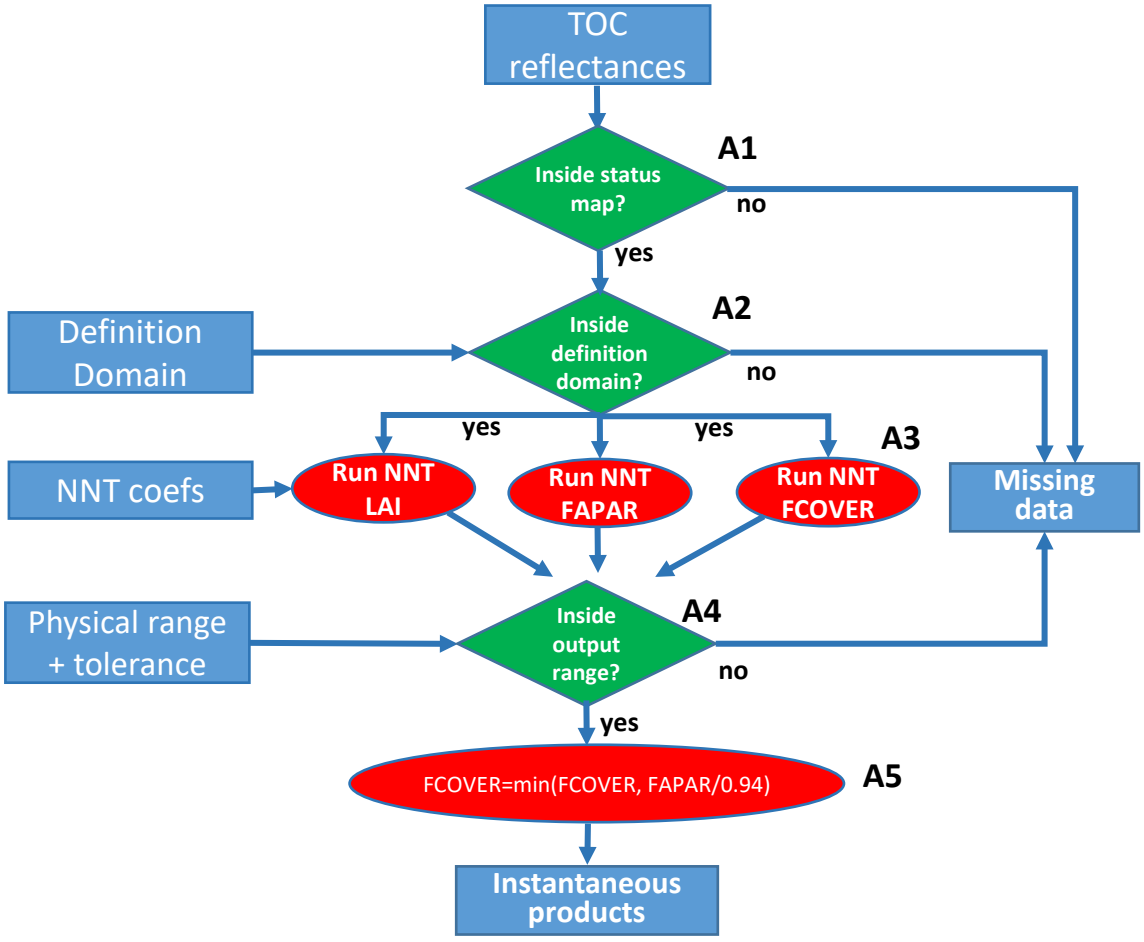


Figure 3: Flow chart describing the instantaneous product estimation process (Step A)

2.5.4.2 Step B: compositing, smoothing and gap filling

The compositing, smoothing and gap filling are run for each dekadal date *D*. Two cases are considered: EBF and nonEBF. The gap filling is only applied for EBF processing. The identification of the class is achieved after running tentatively the compositing module for the EBF class. Note that similarly to Step A, the algorithm produces in parallel the three variables since the final dekadal products are strongly inter-related. This step is sketched in Figure 4.

1. ***Run the Evergreen Broadleaf Forest (EBF) compositing (B1)***. The pixels located in tropical latitudes $(-28.5º \leq Latitude \leq 28.5º)$ or in Australia $(Latitude < 0º$ and $115º \leq Longitude \leq 155º)$ are first processed as EBF. The compositing, temporal smoothing and gap filling is applied to the instantaneous estimates from NNTs (Step A), with a specific set of parameters for EBF case. The composited value results from the mean value of the 10 instantaneous estimates the closest to the dekad D being processed within a maximum

compositing window of 210 days in the semi-period before and 60 days in the period after D. If less than 10 valid instantaneous estimates are available, the product value at date D is set to the previous EBF product value at D-1. The algorithm provides as output the LAI, FAPAR and FCover estimates at dekadal date *D* as well as an assignment to the EBF or nonEBF class. This assignment is derived both from the instantaneous products time series themselves after a regularization procedure using the landcover estimates from the previous 36 dekads. However, at the beginning of PROBA-V time series, when the number of available dekads ($ND_{available}$) is lower than 36 (first year), the CCI-LC 2014 land cover is used as auxiliary information for the regularization. For the regularization of the land cover for Sentinel-3 time series when $ND_{available}<36$, Sentinel-3 landcover estimates are completed with the previous $36-ND_{available}$ landcover estimates from PROBA-V time series. If the pixel is detected as nonEBF, a second run is performed, with specific nonEBF NNTs in step A and step B2 smoothing procedure (Figure 4). For the pixels which are not located in tropical latitudes or Australia, the algorithm is run in one single iteration using nonEBF branch.

2. ***Case of non-Evergreen Broadleaf Forest (B2)***. If the pixel is considered as nonEBF, the nonEBF algorithm is triggered. It first rejects outliers based on the smoothness of the instantaneous products time series over a restricted window. Then, the compositing is applied by fitting a linear or second-degree polynomial depending on the number of available instantaneous observations within a maximum compositing window of ±60 days and a special set of parameters for each specific case. The algorithm provides as output the LAI, FAPAR and FCover estimates at dekadal date *D*.

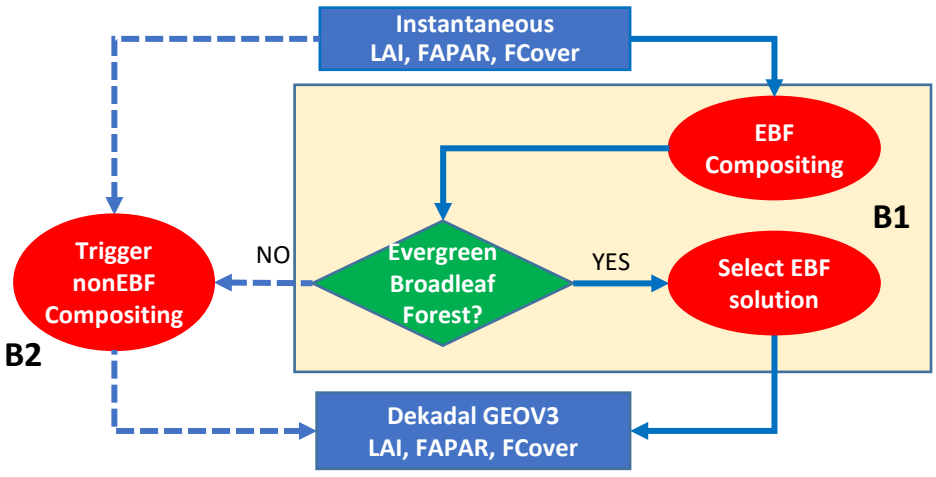


Figure 4: Flow chart describing Step B.

2.5.4.3 Quantitative quality indicators

In addition to the LAI, FAPAR and FCover values, qualitative (see QC indicators in the ATBD, CGLOPS1_ATBD_ LAI[FAPAR/FCOVER]300m-V2.0) and quantitative quality assessment indicators (QA) are also generated:

- Number of available valid instantaneous values in the compositing window (NOBS); the more instantaneous values the more reliable are the products.
- Length in days of semi-period before *D* (LENGTH_BEFORE); the shorter the period, the more reliable are the products.
- Length in days of semi-period after *D* (LENGTH_AFTER); the shorter the period, the more reliable are the products.
- Root Mean Square Error (RMSE) on LAI. It is computed between the final dekadal product value and the available valid instantaneous values in the compositing window.
- Root Mean Square Error (RMSE) on FAPAR. It is computed between the final dekadal FAPAR product value and the available valid instantaneous FAPAR values in the compositing window.
- Root Mean Square Error (RMSE) on FCover. It is computed between the final dekadal FCover product value and the available valid instantaneous FCover values in the compositing window.

2.6 DIFFERENCES WITH PREVIOUS VERSIONS OF 300M PRODUCTS

The main differences of Version 2.0 algorithm with Version 1.0 (PROBA-V products) and Version 1.1 (OLCI products) of LAI, FAPAR, FCover 300m products refers to the input data and its preprocessing, i. e. atmospheric correction:

- Atmospheric correction: Version 1.0, Version 1.1 and Version 2.0 use SMAC method for the atmospheric correction. The differences are that, in Version 1.0, the SMAC code is integrated in LAI/FAPAR/FCover retrieval algorithm which ingests Top of Atmosphere (TOA) reflectance as input data while, in Version 1.1 and Version 2.0, the SMAC correction is applied in an earlier step of the CLMS processing chain and the LAI/FAPAR/FCover retrieval algorithm uses TOC reflectance as input data.
- Input reflectance
    - Instantaneous TOC reflectance products are used in Version 1.1 and Version 2.0 replacing the TOA daily synthesis reflectance used in Version 1.0.
    - Collection 2 of PROBA-V data are used in Version 2.0 while Collection 1 of PROBA-V was used in Version 1.0.
    - OLCI TOC reflectance Version 2.3 are used as input for products Version 2.0 while OLCI TOC reflectance Version 1.0 was used for product Version 1.1 up to 30 June

2022 and OLCI TOC reflectance Version 1.1 was used for product Version 1.1 from 1st July onwards.

- Auxiliary data
  - Atmospheric conditions: Information of the actual water vapor, ozone, pressure (derived from altitude) was required as input data in Version 1.0 to correct the TOA reflectance from atmospheric effects (except aerosol) using the SMAC4.0 model (Rahman and Dedieu 1994). This input data is not used in Version 1.1 and Version 2.0 since the algorithm ingests directly TOC reflectance that were already atmospherically corrected using SMAC method [CGLOPS1_ATBD_S3-AC-V1.1].
  - Land cover: The CCI-LC (https://www.esa-landcover-cci.org) land cover is used in Version 1.1 and Version 2.0 instead of the GLOBCOVER (Defourny et al. 2009) map used in Version 1.0.
- Algorithm parameters: The parameters of the algorithm have been fine-tuned in Version 2.0 using actual PROBA-V and Sentinel-3 OLCI data. In particular, the weights and bias of neural networks (step A) are specific for PROBA-V Collection 2 and Sentinel-3 OLCI since the training datasets (input and target variables) are different between Version 1.1, on one hand, and Version 1.1 and Version 2.0, in other hand. Note that the neural networks and their parameters are exactly the same for Version 1.1 and Version 2.0 Sentinel-3 products. Some parameters of temporal composition (step B) were fine-tuned in Version 2.0.
- Some modifications were also made in the Version 2.0 (and Version 1.1) retrieval algorithm to address the limitations identified in Version 1.0. In particular, a condition was added in Step A for ensuring FAPAR-FCover consistency. And, the post-filtering applied in Step B for RT0 estimates was modified to filter inconsistent estimates.

2.7 LIMITATIONS

The Version 2.0 algorithm for PROBA-V and Sentinel-3 300m products (and Sentinel-3 Version 1.1 300m products) capitalizes on the development and validation of already existing PROBA-V Version 1.0 300m products and the use of neural networks. The PROBA-V Version 1.0 products used as target products for Version 2.0, the original CYCLOPES V3.1 and MODIS C5 used as target products for Version 1.0, the efficacy of the training process and the criteria used to define the input outliers will determine, respectively, the magnitude and range of variation of the final products, their reliability and the definition domain used to remove contaminated values.

The Version 2.0 products are not accurate over the wetlands and can provide unrealistic values over water areas. This can be explained as a legacy effect since the Version 1.0 PROBA-V products used to train the Version 2.0 neural networks do not perform well over such surfaces since the original CYCLOPES (V3.1) and MODIS (C5) products were developed from physical radiative transfer models which are not applicable over heterogeneous surface mixed with water (Xu et al. 2020). In

addition, these particular cases are not included in the training dataset used to generate CLMS 300m products.

Some limitations of 300m products Version 2.0 are associated to the reliability of the CCI-LC v2.0.7cds land cover even this auxiliary information was only used at the beginning of time series processing when no enough dekads are available for the estimation of EBF or nonEBF land cover class. The algorithm must optimally be applied in a sequential order starting with PROBA-V followed by Sentinel-3 processing. In this optimal case, the CCI-LC 2014 is used for the initialization EBF and nonEBF masks at the beginning of PROBA-V processing and for Sentinel-3 time series not external information is required but the EBF/nonEBF class estimates resulted from Version 2.0 algorithm detection for the last dekads of PROBA-V processing are used. Note that because of the compositing period, the first dekad of the Sentinel-3 distributed products should preferentially be ~1 year after the date of the first input data ingested in the algorithm, more precisely at least 210 days after which corresponds to the maximum length of the half compositing window before the date being processed for EBF case.

The consistency across missions of long time series of 300m products highly depends on the temporal consistency of input TOC reflectance between PROBA-V and Sentinel-3 sensors and across the different processing chains (e. g. cloud-aerosol screening).

The efficiency of the temporal methods used in the composition of 300m products from instantaneous estimates depends on the level of noise and gaps in the time series. In this sense, the Collection 2 of PROBA-V data used as input in Version 2.0 is expected to improve the cloud screening of the earlier Collection 1 of PROBA-V used in Version 1.0 (https://proba-v.vgt.vito.be/en/quality/product-and-algorithm-information). The Sentinel-3 TOC reflectance includes a cloud screening based on IDEPIX [CGLOPS1_QAR_S3-CloudMask]. The final product is also dependent on the criteria used to filter the residual outliers since outlier rejection constitutes a critical step in the algorithm.

When used in projection mode for cases where little observations are available, the polynomial fit applied in the temporal smoothing and compositing may lead to inconsistent estimates. The time series of real time estimates (RT0) still shows more noise compared to RT1, RT2 and the consolidated RT6 since for these modes more observations are available.

The approach used to process pixels flagged as EBFs constitutes an oversimplification of the reality because of the possible seasonality of evergreen broadleaf forests. The high uncertainty associated with the data due to poor atmospheric correction and very high cloud occurrence in equatorial and tropical latitudes prevented the extraction of meaningful seasonality at the resolution of the individual pixels.

The user should use the LAI, FAPAR, FCover 300m products with due attention to the quality flags values as well as the associated uncertainties, in particular for areas with long periods of cloudiness.

3 PRODUCT DESCRIPTION

3.1 FILE NAMING

The **LAI, FAPAR, FCover** 300m products follow the following naming standard:

c_gls_<Acronym>[-<RTx>]_<YYYYMMDDHHMM>_<AREA>_<SENSOR>_V<VERSION>

where

- <Acronym> is LAI300, FAPAR300 or FCOVER300, the 300 references that the product is provided at 300m spatial resolution.
- <RTx> is the consolidated period, as explained in 3.4.3:
    - RT0: Near Real Time product
    - RTn: Consolidated Real Time Product, where n equals the number of times the RT0 product was updated.
- <YYYYMMDDHHMM> gives the temporal information of the file. YYYY, MM, DD, HH, and MM denote the year, the month, the day, the hour, and the minutes, respectively.
- <AREA> gives the spatial coverage of the file. In this case, <AREA> is GLOBE, the name of the entire globe.
- <SENSOR> gives the name of the sensor used to retrieve the product, with S3 referring to Sentinel-3 and PV to PROBA-V
- <VERSION> shows the processing line version used to generate these LAI, FAPAR, FCover products. The version denoted as M.m.r (e.g. 1.1.1), with 'M' representing the major version (e.g. V1), 'm' the minor version (starting from 0) and 'r' the production run number (starting from 1) (Table 3).

Table 3: Explanation in version numbering and recommendations for using efficiently the products

Versions	Differences	Recommendations
Major	Significant change to the algorithm.	Do not mix various major versions in the same applications, unless it is otherwise stated.
Minor	Minor changes in the algorithm	Can be mixed in the same applications, but require attention or modest modifications
Run	Fixes to bugs and minor issues. Later run automatically replaces former	Consider it as a drop-in replacement

3.2 FILE FORMAT

The LAI, FAPAR, FCover 300m products are distributed in compressed Network Common Data Form version 4 (netCDF4) files with attributes compliant with version 1.6 of the Climate and Forecast (CF) conventions and contain the following layers:

- LAI (or FAPAR, or FCOVER)
- NOBS: number of instantaneous values available for the compositing (QA1 in Table 5)
- LENGTH_BEFORE: length in days of the semi-period before the dekadal date (D) of the compositing window (QA2 in Table 5)
- LENGTH_AFTER: length in days of the semi-period after the dekadal date (D) of the compositing window (QA3 in Table 5)
- RMSE: root mean square error on LAI (or FAPAR, or FCover) (QA4, QA5, QA6 in Table 5)
- QFLAG: quality flag (Table 6)

3.3 PRODUCT CONTENT

The physical ranges of LAI, FAPAR, FCover are given in Table 4. The physical values are retrieved by:

$$\mathbf{PhyVal = DN * scale_factor + add_offset} \qquad \mathbf{Eq.\ 1}$$

where DN is the digital number in the file and the scaling factor and the offset are given in Table 4. As an example, LAI is a byte for which a data value of 180 equals a physical value of 6.

Table 4: Range of values and scaling factors of LAI, FAPAR and FCover.

	LAI	FAPAR	FCover
Minimum physical value	0	0	0
Maximum physical value	7.0	0.94	1.0
Maximum DN value	210	235	250
Missing value	255	255	255
Scale_factor	1/30.	1/250.	1/250.
Add_offset	0	0	0

In addition to the LAI, FAPAR and FCover, a set of quality indicators are also provided. They are listed in Table 5 and Table 6, respectively. The quality flag (QFLAG) must be read bit per bit. The QFLAG is coded on 1 byte with a nodata value equal to 255.

The water pixels (Bit 1 = 1) are not processed and the product values (Table 4) and quality indicators (Table 5) are set to 255. The QFLAG value of water pixels (Table 6) can have value 1 or 255.

Table 5: Range of values and scaling factors of quantitative quality indicators. *D* means dekadal date.

QA	Quality indicator	Physical Minimum	Physical Maximum	Max DN value	Scale_factor	Add_offset	Missing Value
1	Number of available valid instantaneous estimates in the compositing window (NOBS)	0	60*	60	1	0	255
2	Length in days of semi-period before *D* (LENGTH_BEFORE)	0	210	210	1	0	255
3	Length in days of semi-period after *D* (LENGTH_AFTER)	0	60	60	1	0	255
4	RMSE on LAI	0.0	7.0	210	1/30.	0	255
5	RMSE on FAPAR	0.0	0.94	235	1/250.	0	255
6	RMSE on FCover	0.0	1.0	250	1/250.	0	255

** The theoretical maximum of NOBS is set to 60 to account for more than one instantaneous estimate per day in the compositing window but the typical maximum of NOBS is 10 for EBF and 20 for nonEBF.*

Table 6: Quality flag (QFLAG) of Version 2.0 300m products

Bit	Qualitative quality indicator	Meaning
1	Land/Water	Bit 1 = 0: Land Bit 1 = 1: Water
2	Vegetation class identified	Bit 2 = 0: nonEBF case Bit 2 = 1: EBF case
3	Not used	Not used
4	Not used	Not used
5	Method used for product computation: EBF case	EBF case - only applicable when bit 2=1: Bit 5 = 0: Based on instantaneous observations Bit 5 = 1: Based on previous dekadal product or missing value (in this case the product value is set to missing value)
6:7	Method used for product computation: nonEBF case	NonEBF case – only applicable when bit 2=0: Bits (6:7) = 00: Second degree polynomials fit Bits (6:7) = 01: Linear fit Bits (6:7) = 10: Interpolation between the two nearest dates within a 15-day local window Bits (6:7) = 11: Nearest data within a 5-day window or missing value (in this case the product value is set to missing value)
8	Instantaneous EBF classification flag	Bit 8 = 0: nonEBF case instantaneous value Bit 8 = 1: EBF case instantaneous value

The netCDF files contain a number of netCDF metadata attributes:

- on the file-level (Table 7);
- on layer level (Table 8)
- at the level of the standard dimension variables for latitude (‘lat’) and longitude (‘lon’), holding one value per row or column respectively (Table 9);
- at the level of the grid mapping (spatial reference system) variable (‘crs’) (see Table 10).

Table 7: Description of netCDF file attributes

name	Description	Data Type	Example (LAI)
Conventions	Version of the CF-Conventions used	String	CF-1.6
title	A description of the contents of the file	String	10-daily Leaf Area Index 333M: GLOBE 2020-07-10T00:00:00Z
institution	The name of the institution that produced the product	String	VITO NV
source	The method of production of the original data	String	Derived from EO satellite imagery
history	A global attribute for an audit trail. One line, including date in ISO-8601 format, for each invocation of a program that has modified the dataset.	String	2025-09-22: Processing line GEO
references	Published or web-based references that describe the data or methods used to produce it.	String	http://land.copernicus.eu/global/products/lai
archive_facility	Specifies the name of the institution that archives the product	String	VITO
product_version	Version of the product (VM.m.r)	String	V2.0.1
time_coverage_start	Start date and time of the total coverage of the data for the product. Defined as 0h UTC on the date that is 210 days before *D*, but not before the start of the PROBA-V mission data (2013-10-16).	String	2019-10-23T00:00:00Z
time_coverage_end	End date and time of the total coverage of the data for the product.	String	2020-07-10T23:59:59Z
platform	Name(s) of the orbiting platform(s)	String	SENTINEL3A,SENTINEL3B
sensor	Name(s) of the sensor(s) used	String	OLCI
identifier	Unique identifier for the product	String	urn:cgls:global:lai300_v2_333m:LAI300-RT0_202007100000_GLOBE_S3_V2.0.1
parent_identifier	Identifier of the product collection (time series) for the product in the CLMS metadata catalogue.	String	urn:cgls:global:lai300_v2_333m
long_name	Extended product name	String	Leaf Area Index
orbit_type	Orbit type of the orbiting platform(s)	String	LEO
processing_level	Product processing level	String	L3

name	Description	Data Type	Example (LAI)
processing_mode	Processing mode used when generating the product (Near-Real Time, Consolidated or Reprocessing)	String	Near Real Time
copyright	Text to be used by users when referring to the data source of this product in publications (copyright notice)	String	Copernicus Service information 2026

Table 8: Description of netCDF layer attributes

Attributes	Description	Data Type	Examples for LAI layer
CLASS	Dataset type		
standard_name	A standardized name that references a description of a variable's content in CF-Convention's standard names table. Note that each standard_name has corresponding unit (from Unidata's udunits).	String	leaf_area_index
long_name	A descriptive name that indicates a variable's content. This name is not standardized. Required when a standard name is not available.	String	Leaf Area Index 333M
scale_factor	Multiplication factor for the variable's contents that must be applied in order to obtain the real values. Omit in case the scale is 1.	Float	0.033333
add_offset	Number to be added to the variable's contents (after applying scale_factor) that must be applied in order to obtain the real values. Omit for offset 0.	Float	0.0
units	Units of the variable's content, taken from Unidata's udunits library. Empty or omitted for dimensionless variables. Fractions should be indicated by scale_factor.	String	
valid_range	Smallest and largest values for the variable. Missing data is to be represented by one or several values outside of this range.	byte	0, 210
_FillValue	Single value used to represent missing or undefined data and to pre-fill memory space in case a non-written part of data is read back.	byte	255

	Value must be outside of valid_range.		
missing_value	Single value used to represent missing or undefined data, for applications following older versions of the standards. Value must be outside of valid_range.	byte	255
grid_mapping	Reference to the grid mapping variable	String	crs
flag_values	Provides a list of the flag values. Used in conjunction with flag_meanings.	byte	255
flag_meanings	Descriptive phrases for each flag value, separated by blanks	String (blank separated list)	Missing

Table 9: Description of netCDF attributes for coordinate dimensions (latitudes and longitudes)

Attribute	Description	Data Type	Example
CLASS	Dataset type	String	DIMENSION_SCALE
DIMENSION_LABELS	Label used in netCDF4 library	String	lon
NAME	Short name	String	lon
standard_name	A standardized name that references a description of a variable's content in CF-Convention's standard names table. Note that each standard_name has corresponding unit (from Unidata's udunits).	String	longitude
long_name	A descriptive name that indicates a variable's content. This name is not standardized. Required when a standard name is not available.	String	longitude
units	Units of the variable's content, taken from Unidata's udunits library.	String	degrees_east
axis	Identifies latitude, longitude, vertical, or time axes.	String	X
_CoordinateAxisType	Label used in GDAL library	String	Lon

Table 10: Description of netCDF attributes for the grid mapping variable

Attribute	Description	Data Type	Example
GeoTransform	Six coefficients for the affine transformation from pixel/line space to coordinate space, as defined in GDAL's GeoTransform	String	-180.0000000000 0.0029761905 0.0 80.0000000000 0.0 -0.0029761905
_CoordinateAxisTypes	Label used in GDAL library	String, blank separated	GeoX GeoY
_CoordinateTransformType	Type of transformation	String	Projection
grid_mapping_name	Name used to identify the grid mapping	String	latitude_longitude
inverse_flattening	Used to specify the inverse flattening (1/f) of the ellipsoidal figure associated with the geodetic datum and used to approximate the shape of the Earth	Float	298.257223563
long_name	A descriptive name that indicates a variable's content.	String	coordinate reference system
longitude_of_prime_meridian	Specifies the longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum	Float	0.0
semi_major_axis	Specifies the length, in metres, of the semi-major axis of the ellipsoidal figure associated with the geodetic datum and used to approximate the shape of the Earth	Float	6378137.0
spatial_ref	Spatial reference system in OGC's Well-Known Text (WKT) format	String	GEOGCS["WGS 84",DATUM["WGS_1984",… AUTHORITY["EPSG","4326"]]

3.4 PRODUCT CHARACTERISTICS

3.4.1 Projection and Grid Information

The product is displayed in a regular latitude/longitude grid (plate carrée) with the ellipsoid WGS 1984 (Terrestrial radius=6378km). The resolution of the grid is 1/336°. The reference is the centre of the pixel (integer degrees are in the centre of a pixel).

3.4.2 Spatial information

The Sentinel 3 derived products are provided from longitude 180°W (at centre of pixel) to 180°E and latitude 80°N to 65°S.

The Proba-V derived products have only valid data between latitude 75°N and 65°S because of the sensor range.

3.4.3 Temporal Information

The LAI, FAPAR, FCover 300m are near real time products, updated every 10 days within a consolidated period. The temporal information "YYYYMMDDHHMM" in the filename is the "**nominal**" or reference date corresponding to the product value, (D), the last day of the dekad (10-day period). The RT0 product is updated with two consolidation periods (RT1 and RT2) and a final consolidation (RT6), as shown in Figure 5.

For example, the file c_gls_LAI-RT0_202309100000_GLOBE_S3_V2.0.1.nc corresponds to the near real time LAI product of first dekad (D) of September 2023. The product is generated using the past observations till D within the compositing window.

For example, the file c_gls_LAI-RT2_202309100000_GLOBE_S3_V2.0.1.nc corresponds to the consolidated LAI product two dekads (D+2) after the first dekad of September 2023. The actual product date still corresponds to the first dekad of September 2023. However, the product is generated using the past observations till D within the compositing window plus the observations of two dekads in the consolidation period (between D and D+2).

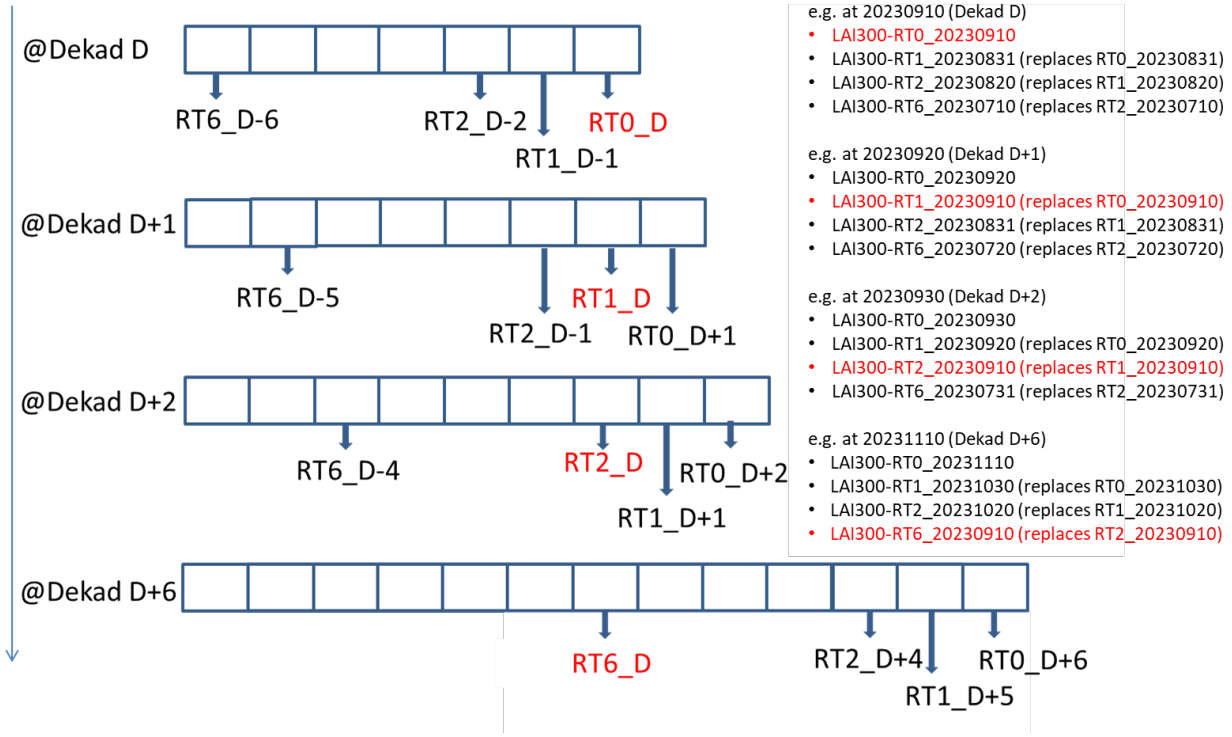


Figure 5 : Consolidation period example

As the product value after the second consolidation remains stable, it has been decided not to distribute the 3rd, 4th and 5th consolidation (RT3, RT4 and RT5), but only a final (RT6) consolidation taking benefit of the full 60 days LENGTH_AFTER (Table 5).

For reprocessed historical data, only the most accurate consolidation RT6 is produced.

3.5 DATA POLICIES

The Copernicus component is governed by Regulation (EU) No 2021/696 of the European Parliament and of the Council of 28 April 2021 establishing the Union Space Programme and the European Union Agency for the Space Programme.

The Copernicus land monitoring products and services are made available on a principle of **full, open and free access**, as established by the Commission Delegated Regulation (EU) No 1159/2013 of 12 July 2013.

How to cite and reference the products and services of the Copernicus Land Monitoring Service is explained here: https://land.copernicus.eu/en/data-policy.

3.6 ACCESS AND CONTACTS

The LAI, FAPAR, FCover 300m products are available through the Copernicus Land Monitoring Service website at the address: https://land.copernicus.eu/ with contact information (helpdesk) on https://land.copernicus.eu/en/contact-service-helpdesk.

4 SUMMARY OF VALIDATION EXERCISE

4.1 VALIDATION PROCEDURE

The quality assessment of Sentinel-3 OLCI LAI, FAPAR and FCover V2.0 products is conducted according the methodology described in the Product Quality Assurance Document [CGLOPS1_PQAD], in agreement with the Committee on Earth Observation Satellite (CEOS) Land Product Validation (LPV) best practices for validation of LAI products (Fernandes et al., 2014). Two main validation approaches are defined: direct validation (i.e., comparison of satellite products with in situ measurements) and indirect validation (or product intercomparison). Several criteria of performance are evaluated, including completeness, spatial consistency, temporal consistency, error evaluation (Accuracy, Precision and Uncertainty) and conformity test. For the conformity test, two uncertainty predefined levels (optimal, target) are defined according to CGLOPS requirements (see Annex).

4.2 MAIN RESULTS

The validation is ongoing.

5 PRODUCT USAGE

At the time of writing this document, we found few studies using the CLMS LAI, FAPAR and FCover 300m products as the temporal series covers only 12 years (2014-present), they were validated quite recently and the reprocessing with Version 2.0 algorithm to improve the consistency of time series and correct the geolocation issues in Sentinel-3 products was not completed. A part from the known issues of geolocation that motivated the reprocessing, any known issues have been reported. Usages are similar to previous products including data analysis and data assimilation into land surface models as we describe in the following subsections. Few studies already demonstrated the suitability of CLMS 300m products for these two specific application fields. Radočaj (2025) took explicitly advantage of the 300m resolution of CLMS LAI and FAPAR products to successfully predict FAO cropland suitability for major crops in Europe. Heyvaert et al. (2024) evaluated data assimilation of three satellite products (microwave surface soil moisture product, vegetation optical depth product, and LAI product) into a land surface model and showed that the largest improvements in estimates of evapotranspiration, gross primary production and net ecosystem were observed when the CLMS 300m LAI product was assimilated.

5.1 ANALYSIS

The LAI can be used for detection of change and for providing information on shifting trends or trajectories in land use and cover change.
FCover is a very useful and understandable product since it may be directly compared with the human perception. Therefore, its temporal evolution can be directly linked to environmental applications such as deforestation and other types of land cover changes. FCover is relevant for a wide range of Land Biosphere Applications such as agriculture and forestry, environmental management, hydrology, natural hazards monitoring and management, and fire scar extent, among others. The product is used as an input for the start of the growing season. It can also be used for year-to-year comparison of vegetation status.
FAPAR can be used into light use efficiency models at the daily time step, assuming that the radiation is coming mainly from the sun directions (Monteith, 1972). It can be related to the biomass production of crops or grassland (Kowalik et al, 2014; Diouf et al., 2015).

5.2 ASSIMILATION IN MODELS

The green LAI, which is the main vegetation structure variable governing canopy reflectance, is also one of the main drivers for canopy functioning processes (Jacquemoud and Baret, 1990; Jacquemoud et al., 2009). LAI represents the size of the interface between the plant and the atmosphere for energy and mass exchanges. It is thus of prime interest for energy balance, photosynthesis, transpiration and litter production (Barbu et al., 2014; Albergel et al., 2018). LAI could be used to validate canopy

photosynthesis models which simulate growth and canopy development based on climate and environmental factors (Warnant et al., 1994; Kohlmaier et al., 1997; Kergoat, 1998). It is also a sensitive parameter for the control of evapotranspiration in Soil-Vegetation-Atmosphere-Transfer (SVAT) schemes within the context of General Circulation Models (Mahfouf et al., 1995; Neilson and Drapek, 1998) and Numerical Weather Prediction models (Boussetta et al. 2013).
Similarly, the FCover is a crucial parameter of SVAT models to separate the contribution of vegetation and soil in the surface energy balance and evapotranspiration computation (Allen, 2000; Aouade et al., 2020).
FAPAR is essential in production efficiency models to estimate carbon fluxes (McCallum et al. 2009), among many other applications.
In this sense, the assimilation of LAI, FAPAR and FCover into Land Surface Models may be used to improve the characterization of the annual cycle of the hydrological (van den Hurk et al. 2003) and carbon fluxes in the global climate-carbon cycle.

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7 ANNEX: REQUIREMENTS

According to the applicable document [AD1], the requirements relevant for LAI, FAPAR, FCover products are described below.

7.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

PRODUCT SPECIFICATION	
Geometric Properties:	
Baseline dataset pixel resolution	300m
Target baseline location accuracy	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetical datum	WGS84
Geographic projection	Regular latitude/longitude grid
Geographic coverage:	Global
Temporal resolution	10-day period (dekad: days 1-10, 11-20, 21 end of month)
Timeliness	Within 2 days (optimally 1 day) after the end of each dekad

FAPAR – FRACTION OF ABSORBED PHOTOSYNTHETICALLY ACTIVE RADIATION	
FAPAR corresponds to the fraction of photosynthetically active radiation (PAR, i.e. solar radiation in the spectral range 400-700 nm) that is absorbed by plants.	
Uncertainty (2-sigma)	Threshold: 10% for values >0.5; 0.005 (absolute values) for smaller values Goal: 5% for values ≥ 0.05; 0.0025 (absolute value) for smaller values
Stability (per decade i.e. 10 years)	Threshold: 3% Goal: 1.5%

LAI – LEAF AREA INDEX	
Leaf Area Index of a plant canopy or ecosystem is defined as one half of the total green leaf area per unit horizontal ground surface area and measures the area of leaf material present in the specified environment (projection to the underlying ground along the normal to the slope) [m^2/m^2].	
Uncertainty (2-sigma)	Threshold: 20% for values ≥ 0.5; 0.1 (absolute value) for smaller values Goal: 10% for values ≥ 0.5; 0.05 (absolute value) for smaller values
Stability (per decade, i.e. 10 years)	Threshold: 3% Goal: 1.5%

FCOVER – FRACTION OF GREEN VEGETATION COVER	
The Fraction of Vegetation Cover corresponds to the fraction of ground covered by green vegetation. Practically, it quantifies the spatial extent of the vegetation.	
Uncertainty (2-sigma)	Threshold: 10% Goal: 5%
Stability (per decade, i.e. 10 years)	Threshold: 5% Goal: 2.5%

7.2 FURTHER REQUIREMENTS

7.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

7.2.2 Data structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.
Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

7.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets will be available in NETCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the Contracting Authority during the Framework Contract.

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

7.2.4 Uncertainties and validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Level 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL data sets and/or Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component of Copernicus Global Land component when biophysical parameters are available.

7.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land Component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

LST and SWI can be based on geostationary and other satellite data.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary data sets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

7.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination. The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[2] https://land.copernicus.eu/en/products/global-image-mosaic